

ABSTRACT

Radiation monitoring and mapping are fundamental to nuclear safety, environmental monitoring and emergency response. This Summer Internship Report presents the research and development work carried out on the project "**RADMAP – RADiation MAPing – GPS Enabled Radiation Mapping Prototype on Raspberry Pi-5 (3D Visualization)**" at the Amity Institute of Nuclear Science & Technology (AINST). The first goal of the project was to build a portable and intelligent radiation mapping system that would provide real time radiation measurements and geographical coordinates, environmental parameters and current advanced visualization for effective radiation monitoring and future source localization.

During the 47 day internship, a wide range of research and development projects were conducted on embedded systems, geospatial mapping, radiation detection and data visualization. This work included creating and preprocessing a radiation dataset, uploading the data to Kaggle, and using the data to create interactive 2D radiation maps. A real time GPS acquisition module was designed using **GlobalSat BU-353N** GPS receiver, which was attached to the Raspberry Pi-5 to acquire latitude, longitude, UTC time, temperature and humidity data and were used for continuous environmental monitoring and data processing.

The project also included integration and testing of a **CNSPEC Multi-Channel Analyzer (Gamma Spectrometer)** with standard radioactive sources like **Cobalt-60 (Co-60)** and **Cesium (Cs-137)**. Radiation count rates, spectral peaks and detector responses were measured and analyzed with success and dynamic radiation intensity maps were generated. In order to address the limitation of the conventional **2D heatmaps**, research was carried out for radioactive source localization on the **3D Digital Twin** technology. To visualize the monitored environment in real time, a real-time visualization system was developed by synchronizing the live image of the camera with GPS coordinates and radiation data, so that the three-dimensional image of the environment monitored by the camera could be obtained.

Moreover, a full backend framework was created to allow for integration of sensor communication, *GPS telemetry*, radiation processing, and visualization modules into a single system. A color grading algorithm, based on the technology used in night vision imaging, was used to illustrate different radiation intensity levels by using different color scales in the live camera view, and thus provide intuitive radiation hotspot detection. Additionally, a

prototype model in the form of a 3D printable model was designed with the final hardware architecture in mind for future prototype development of **RADMAP**.

Success in implementing and validating the prototype of **RADMAP** confirmed the possibility of implementing radiation mapping in real time based on GPS navigation, embedded computing, and intelligent visualization methods. **Dr. Abhishek Yadav & Prof. (Dr.) Alpana Goel, Director of AINST**, appreciated and approved the project for further development towards next generation radiation mapping, autonomous source localization and advanced Digital Twin technologies. The internship was a great practical experience with embedded systems, IoT, geospatial analytics, radiation instrumentation and research-oriented software development, and helped to develop innovative solutions in the areas of nuclear safety and environmental monitoring.

INTRODUCTION

2.1 Background

Radiation monitoring is a key field of safety and protection in nuclear power, the environment, medical treatment, in industry, and in emergency response. The traditional method of radiation monitoring has been to either measure radiation at a set location or use handheld equipment to view radiation from a distance. In most cases, it is hard to visualize the radiation distribution over a large geographical area in real-time using the traditional radiation monitoring methods. These traditional approaches do not have the ability to precisely locate radioactive sources or offer much assistance to quick response in the event of a radiological emergency.

Intelligent radiation mapping systems are becoming more and more significant, because of the powerful development of **embedded systems, GPS technology, Internet of Things (IoT) and geospatial visualization** techniques. Integrating radiation detectors with GPS modules and edge computing platforms such as Raspberry Pi 5 enables the collection of radiation measurements together with precise geographical coordinates. These datasets can be used to create interactive heat maps, which will help you gain insights into the distribution of radiation in an area being monitored.

To overcome these challenges, the **RADMAP (RADiation MAPing)** project was developed to develop a prototype portable GPS supported radiation mapping system that can record real-time radiation data, environmental parameters and geographical information. The system connects hardware sensors and software visualization tools to ensure the accurate, scalable, and intelligent monitoring of radiation, which can be applied to nuclear safety in the future. By Integration all the other components we need to test and validate each and every particular phase of working of our device to detect gather the data and other functionality.

2.2 Project Motivation

The main motivation for this project was kind of to get past the limits of normal 2D radiation heatmaps. Sure, heatmaps do show the radiation intensity across different spots but they do

not really tell you where the radioactive source is in physical space. Finding the exact source, that is, the origin of radiation is one of the biggest challenges that show up in radiation monitoring systems.

To tackle this research issue , the project basically aimed at pushing conventional 2D radiation mapping into a 3D Digital Twin kind of setup, so it can handle future work on radioactive source localization. The motivation got even stronger because we needed to stitch together real time GPS coordinates, some environmental sensing data, camera feedback, and radiation measurements into one single visualization platform, even if parts of it feel kind of messy at first.

The internship also sort of gave me a chance to learn in a practical way embedded system development, plus Raspberry Pi programming , and GPS telemetry, radiation instrumentation, data visualization, backend development, and even research driven engineering. The whole project, it mixes software creation with real world nuclear technology use cases, which kind of helps move things forward for more intelligent radiation monitoring and environmental surveillance.

2.3 Project Scope

So the scope of the **RADMAP** project kind of includes the whole development of a GPS enabled radiation mapping setup using a **Raspberry Pi-5** and several integrated hardware parts. Basically, the project went through building structured radiation datasets, then making interactive **2D radiation heatmaps**, and at the same time plugging in the **GlobalSat BU-353N** GPS Receiver so it can capture real-time latitude longitude temperature humidity plus a timestamp. After that, all the gathered info got synced with the Raspberry Pi-5 which i guess ties everything together.

The project also covers the integration of a **Gamma Spectrometer (CNSPEC MCA)** for radiation sensing and also some validation, using familiar radioactive references like **Cobalt-60 (Co-60)**, and **Cesium-137 (Cs-137)**. From the radiation readouts that were collected, dynamic heatmaps showing radiation counts, peak intensity and even the spatial pattern were successfully made.

Another big part of this project, is the research and development of a 3D Digital Twin meant for future radioactive source localization. The whole system brings together live camera feedback, **GPS telemetry**, and radiation measurements so it can show a real time three dimensional view. Also, we built a backend architecture that kind a coordinates sensor talking, the environmental context data, the radiation processing, and then the visualization modules. Beside this, a kind of night vision style color grading method was also implemented, to show different radiation intensity levels directly on the live camera feed, which improves situational awareness when people do radiation monitoring.

The outcomes of this project really lay down a solid base for later upgrades, like AI-assisted source localization, plus **LiDAR SLAM integration**, and autonomous robotic navigation. There's also drone-based radiation mapping, and a cloud powered radiation monitoring system, for nuclear safety use-cases. In other words the whole thing becomes a stepping stone, for more advanced and maybe a bit more adaptive monitoring setups later on.

2.4 My Role in the Project

During the internship I sort of actively contributed to both the research and development stages of the RADMAP project, like it was not only one thing. I was involved in developing radiation datasets and getting them published on **Kaggle**. I also worked on designing interactive **2D radiation heatmaps**, and then I integrated the **GlobalSat BU-353N GPS receiver BU-353N** with a **Raspberry Pi-5** for real time telemetry acquisition. Besides that I processed environmental parameters such as latitude longitude temperature and humidity, you know, the usual signals that matter.

I also worked on the integration a validating of the Gamma Spectrometer, using Co-60, and Cs-137 radioactive sources, and while doing that I helped with verification routines. I developed the backend architecture for data acquisition and visualization a bit of it at least, then I researched and implemented the idea of a 3D digital twin for radiation source localization. On top of that, I designed a real-time color grading system, inspired by night-vision imaging and created a 3D printable prototype model of the RADMAP system as well. Overall, these contributions improved the prototype functionality a lot, and they supported its approval for more funding, plus deeper research work coming next.

REQUIRMENT SPECIFIC / PRELIMINARIES

3.1 Existing System

Traditional radiation monitoring setups mainly depend on handheld radiation detectors or fixed monitoring stations, to register radiation levels. Even if these methods give accurate radiation readings, they still don't really provide a real-time geospatial picture, or any steady environmental tracking. In practice, the radiation information is often gathered by hand, and then processed on its own schedule. Because of that, it becomes hard to pinpoint the exact spot of radioactive sources during on-site field work. Also, the typical approach does not neatly bring together radiation sensors, GPS modules, environmental sensors, and the visualization platform, so the whole system ends up less effective when you need broad-area radiation surveillance or fast emergency response.

3.2 Problem Statement

The biggest problem in the radiation monitoring systems we already have is basically that they cannot show the radiation distribution in real time while still, accurately pin pointing where the source is. The standard 2D radiation heatmaps tend to only give radiation intensity across a surface, but they do not really add depth, or the spatial context you would want in the field. Also there is no real integrated **GPS telemetry**, no environmental sensing, and no live visualization, so during nuclear checks or emergency events, it becomes harder to make the right decision quickly. So what is needed is a smarter setup that can do real-time radiation charting, keep an eye on the environment, and later also support localization of the radioactive source.

3.3 Proposed Solution

The proposed solution looks like **RADMAP**, a kind GPS-enabled radiation mapping prototype running on a **Raspberry Pi-5**, it's meant to be practical but also a bit experimental. In the setup, the system pulls together a Gamma Spectrometer, a **GlobalSat BU-353N GPS receiver**, a **DHT22 environmental sensor**, and a camera module, so it can grab radiation counts plus GPS coordinates, along with temperature and humidity and even live video at the same time, kind of altogether.

After that, the captured information goes into a Python-based backend where it's processed into interactive **2D radiation heatmaps**, and also **3D Digital Twin**, for clearer visualization, and for helping with radioactive source localization later on. For the live feed, a color-grading algorithm, inspired by night-vision techniques, is used to show different radiation intensity levels right on the camera view, so it becomes easier to spot radiation hotspots without too much guessing. The architecture is modular too, meaning the design can expand later with AI-based '**SLAM**' **source localization**, **LiDAR integration**, autonomous robotic navigation, and cloud-based radiation monitoring too, if needed.

4.4 Requirement Specifications

Component	Specification
Raspberry Pi 5	8 GB RAM
GlobalSat BU-353N	GPS Receiver
CNSPEC MCA	Gamma Spectrometer
DHT22	Temperature & Humidity Sensor
Camera Module	Live Monitoring

Table 1: Hardware Requirements

Software	Purpose
Raspberry Pi OS	Operating System
Python 3.x	Backend Development
Streamlit	Dashboard
OpenCV	Camera Processing
Plotly	3D Visualization
Folium	2D Heatmap
Pandas	Data Processing
NumPy	Numerical Computing
GitHub	Version Control (VC)

Table 2: Software Requirements

TASK ACCOMPLISHED IN ORGANIZATION

4.1 Development of Radiation Dataset

So, the first phase of the project went into building a more structured radiation dataset, to help with visualization plus analysis stuff. Basically, we took the radiation readings along with GPS coordinates and environmental parameters, also the timestamps, then put it all into a standardized layout. After that, the dataset got published on **Kaggle**, so people could use it for research, testing, and later model development. In practice, this dataset became the base layer for producing radiation heatmaps and for checking whether the visualization algorithms were working or not.

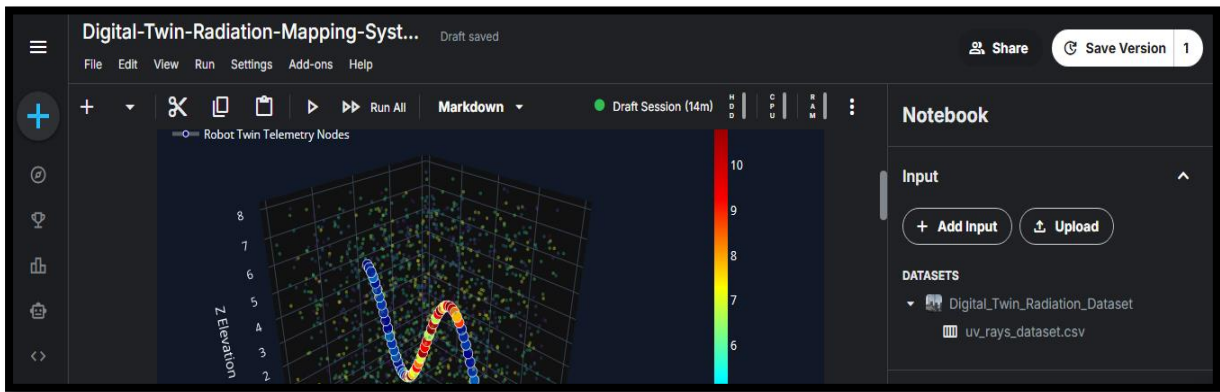


Figure 1: Kaggle Dataset (Dataset page or CSV preview)

4.2 Development of 2D Radiation Heatmap

So, a two dimensional (2D) **radiation heatmap** was developed, basically to show how the radiation intensity ends up distributed over different geographical places. The whole system used radiation readings that came from a Gamma Spectrometer, and then matched those measurements with GPS positions in real time from the **GlobalSat BU-353N GPS Receiver**. Each single radiation value was then placed onto the geographical point it belonged to, so that the radiation levels across the observed region could be visualized accurately, or at least in a way that looks consistent and clear.

The generated heatmap basically showed radiation intensity with a color gradient, where the lower radiation levels were shown in cooler colors, like blue green. While the higher radiation levels showed up in warmer colors, such as yellow, orange, and red. In general this kind of picture made it a bit easier to spot the radiation hotspots and also look at how radiation changes around the surveyed area, in a spatial way, over distance.

To improve the analysis, we also included a few additional factors like the radiation count rate, the peak channel, GPS coordinates, a timestamp, plus temperature and humidity. These inputs were mapped together so they sort of gave a more holistic snapshot of the environmental state next to the radiation readings. In the end, this helped the visualization feel more accurate and reliable, rather than just relying on radiation alone.

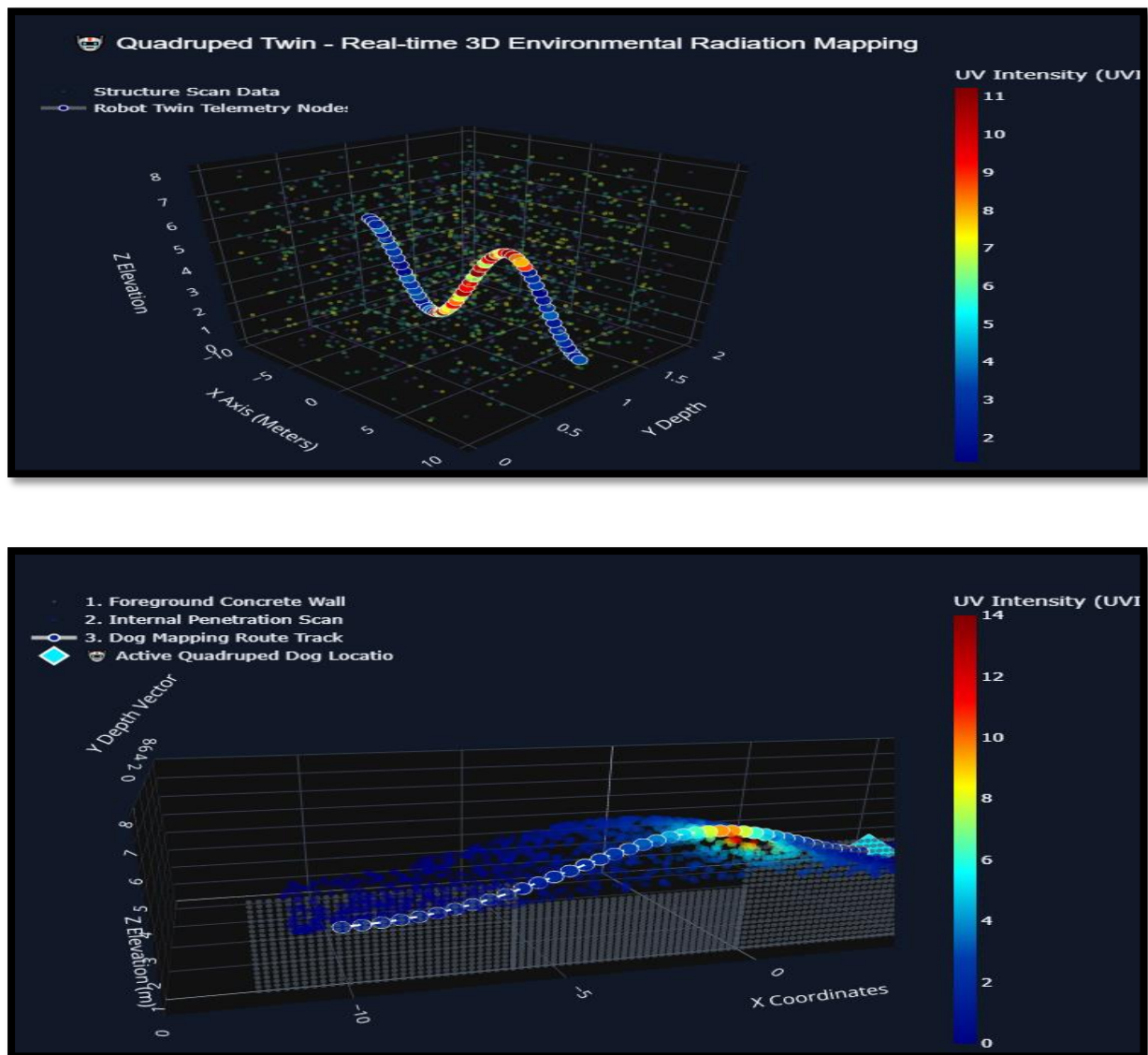


Figure 2: Static 2D Radiation Heatmap Output Generation with Dataset

4.3 GPS Integration with Raspberry Pi-5

Getting **RADMAP** to work well, honestly it needed pretty exact geographic positioning to map the radiation readings right. So, the **GlobalSat “BU-353N”** USB GPS Receiver was connected into the Raspberry Pi-5 so it could grab live location data straight from the GPS satellites. The GPS unit then talks to the Raspberry Pi over the USB link and it keeps sending **NMEA, (National Marine Electronics Association)** sentences that carry the key navigation details, all the time.

We put together a Python based data acquisition module, kind of to pull out and later sort GPS details like **latitude, longitude, altitude, UTC time, speed, satellite condition**, and how accurate the position is. At the same time, beside the GPS telemetry, we also grabbed environment numbers such as temperature and humidity using the DHT22 sensor and then synced all of that with the location data. It means during radiation monitoring, we could collect the geographical info and the weather-is context together, more or less in parallel.

The collected GPS coordinates were pretty much tied to the radiation counts that came in from the Gamma Spectrometer, so each time we measured radiation it also got mapped to the exact location. In a sense, that synced dataset, became the foundation for making the interactive **2D radiation heatmaps** and later it also turned into the input for a **3D Digital Twin visualization system**.

Special focus was placed on real-time communication, and on keeping the data in lockstep so that the time delay between the sensor readings and the GPS updates stayed as low as possible. Python based serial communication tools were used to keep watching the incoming **GPS stream**, then to parse the **NMEA messages**, and finally to save the processed results into organized data structures for more digging, analysis and visualization. Overall, the combined setup delivered a trustworthy telemetry framework, one that can help future mobile radiation monitoring efforts and even support autonomous navigation style applications.

The successful integration of GPS tech has, like, really boosted the **RADMAP** system’s overall ability. It allows for real time geospatial radiation mapping, better environmental record keeping, plus smooth synchronization with the backend visualization platform. In practice it means the platform can track locations more precisely; capture data as it happens

and then line up those updates with the visualization side, without the usual delay or awkward gaps.

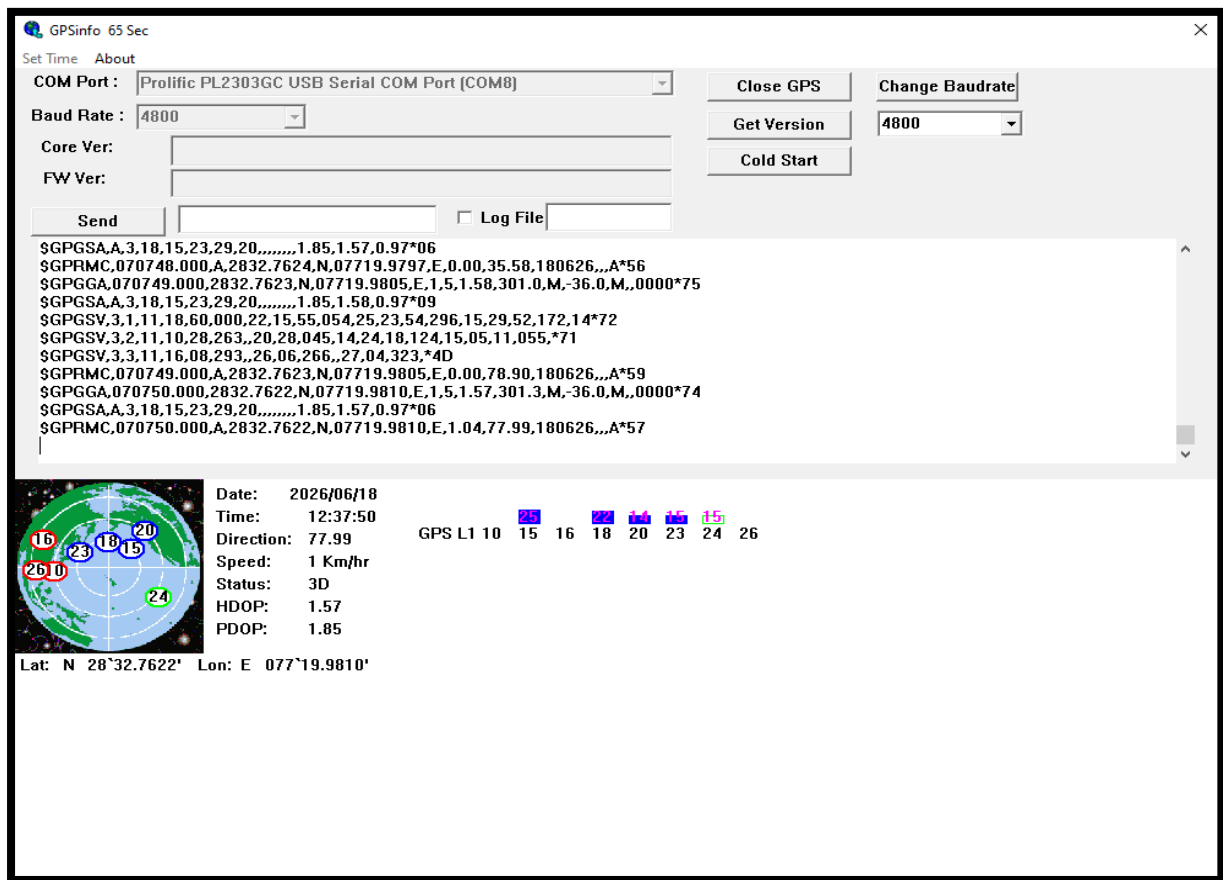


Figure 3: Live GPS Data Terminal Output

4.4 Gamma Spectrometer Integration and Radiation Validation

A critical component in the RADMAP setup is the CNSPEC Multi-Channel Analyser, basically a Gamma Spectrometer, and it was put in place so it could do real time gamma detection along with spectral analysis. This Gamma Spectrometer is used to measure energy and also the intensity of gamma radiation that comes from radioactive materials, in a way that makes it possible to get a solid identification and a more exact, quantitative analysis of the radiation level.

While doing the internship, the spectrometer got configured and sort of interfaced with the Raspberry Pi based backend, for on-going acquisition of radiation measurements. The

detector itself was validated in practice with certified radioactive calibration sources, including **Cobalt-60 (Co-60)** and **Cesium-137 (Cs-137)** . These sources were picked due to their pretty clear gamma energy peaks, and that made them good for calibration tasks, and also for performance checking.

From the radiation measurements that were collected, a few Python visualization modules were crafted to show radiation count rates, peak channels, intensity graphs, and assorted statistical indicators. After that, the processed measurements were also synchronized with **GPS coordinates**, so location aware radiation maps could be generated, and in a way they became more meaningful for real world referencing.

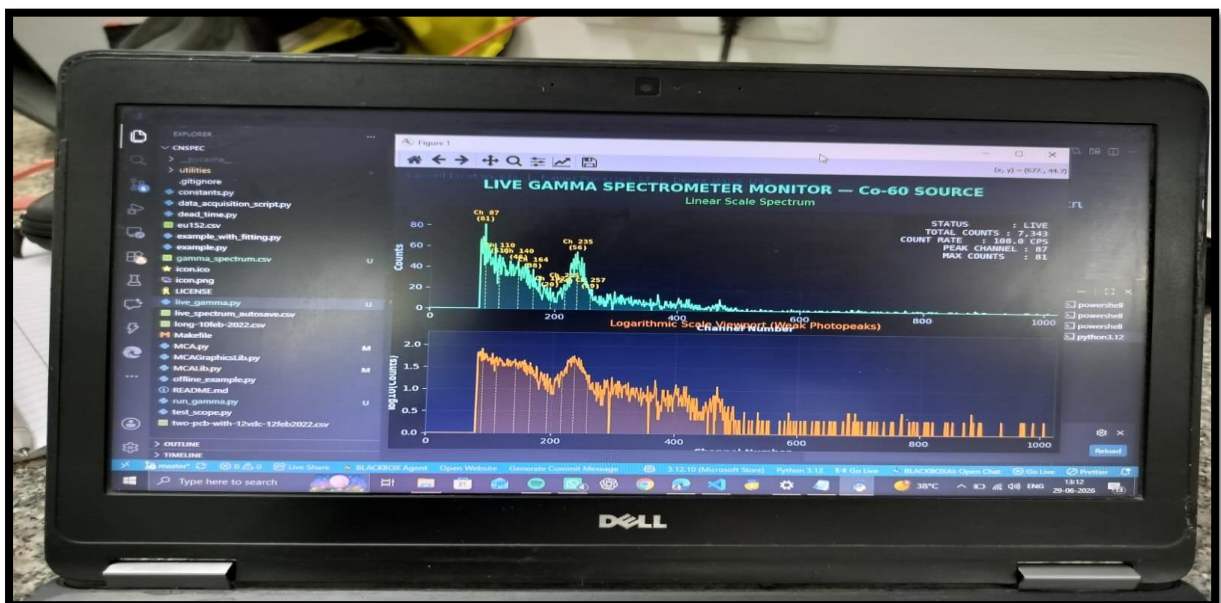
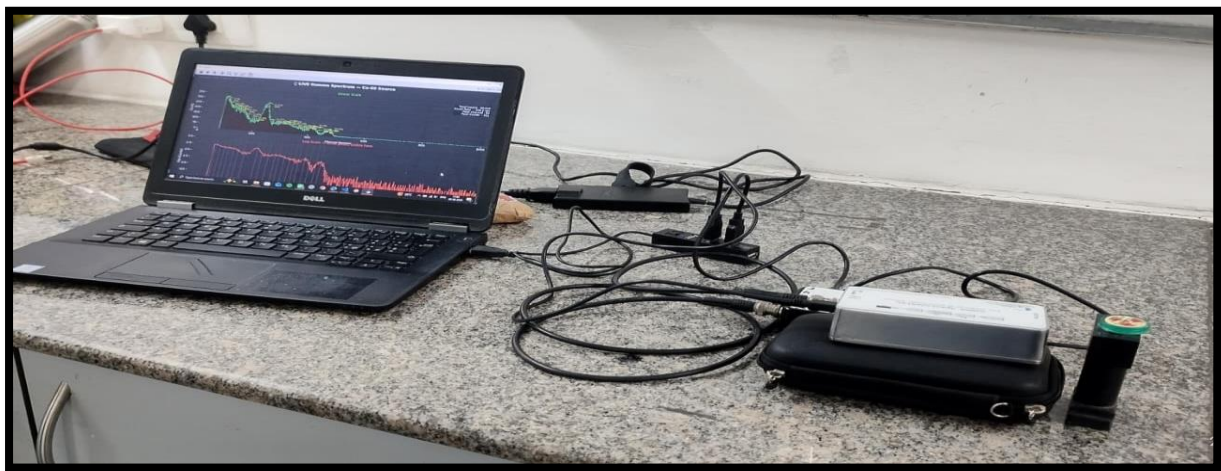


Figure 4: Gamma Spectrometer Setup, Co-60 and Cs-137 Radiation Testing

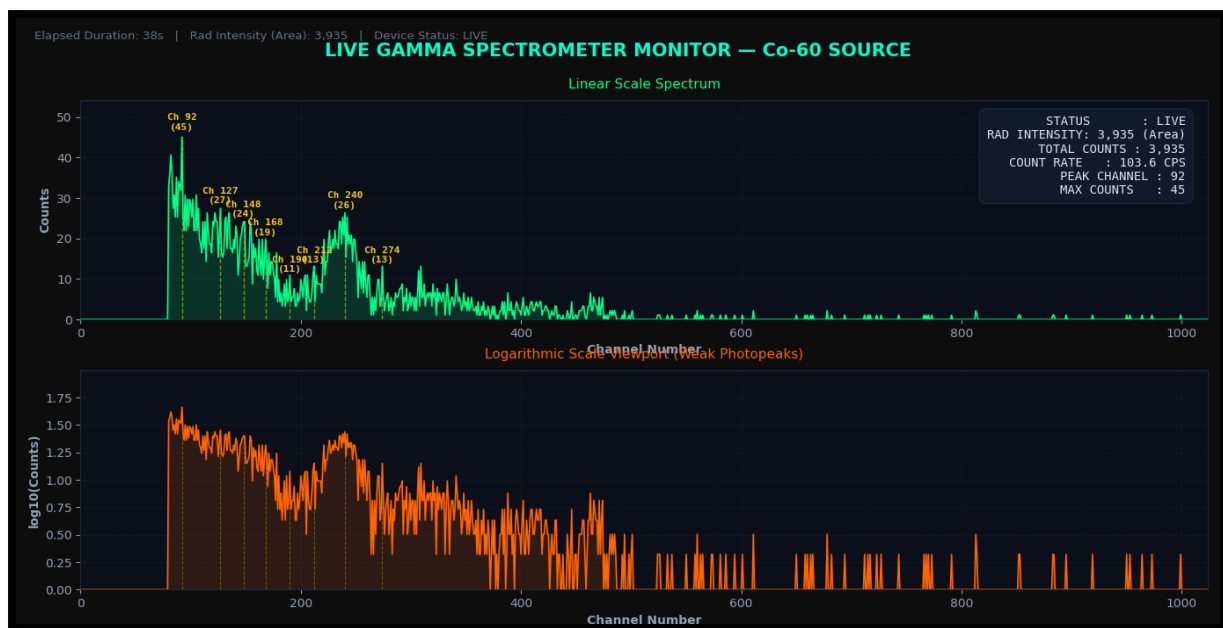


Figure 5: Gamma Spectrum / Peak Detection Graph with Metrics and Radiation Intensity 2D Dynamic Heatmap Generation on real-time basis.

The integration of the Gamma Spectrometer sort of established a solid, continuous radiation tracking base, and it also gave dependable input data for the 2D heat map creation and the **3D Digital Twin display**. In a way the validation showed that the system, as it was built, can accurately spot radiation levels, while still keeping steady real time operation, which is kind of the point anyway.

4.5 Development of 3D Digital Twin

The first **RADMAP** implementation managed to create 2D radiation heatmaps that kind of really showed the radiation intensity across various geographical spots. During later system evaluation though, it was noticed that the usual heatmaps only give a sense of how the radiation levels are spread, but they do not really tell you anything about the likely origin or the actual spatial location of the radioactive source. Because of this, the research and development effort went toward building a “**3D Digital Twin**” that can offer improved spatial visualization, in a more meaningful way.

A Digital Twin is basically a virtual stand-in for a real physical environment; it keeps taking in real-time sensor feeds and, somehow, ends up mirroring the changes that happen in the

area being watched. In this project, we put together a three dimensional visualization framework, by **stitching GPS coordinates, gamma radiation readings, environmental sensor signals, and live camera feedback** into one unified virtual space.

On the backend side, everything runs continuously, syncing the radiation intensity with the camera frames and the matching geographical location, so the monitored region looks alive and changes as expected. And yeah, unlike traditional heatmaps, this **3D Digital Twin** lets users explore the radiation distribution inside a three dimensional spatial context, which makes environmental interpretation much easier and boosts situational awareness, in a more immediate way than before.

To further improve visualization, we designed a night-vision-inspired color grading approach where various radiation intensity bands get shown through several color gradients. The lower radiation areas were rendered with cooler hues, while the higher intensity zones used warmer colors, so that radiation hotspots can be spotted immediately during live monitoring, even when things are fast.

The Digital Twin setup was planned to be modular and scalable so that, later on, we can plug in LiDAR sensing, **SLAM (Simultaneous Localization and Mapping)**, AI-based radioactive source localization, autonomous robotic navigation, and drone-supported radiation mapping. In general, this work should act as the base layer for the next era of intelligent radiation monitoring systems.



Figure 6: 3D Digital Twin Visualization

A real-time color grading algorithm inspired by **night vision imaging technology** was designed to represent varying radiation intensity levels. Different colors were assigned according to measured radiation counts, allowing intuitive identification of radiation hotspots directly on the live camera feed. This approach significantly improves visualization and situational awareness during radiation monitoring. SLAM Model Equation given here:

$$X_t = f(X_{t-1}, U_t, Z_t)$$

4.6 Backend System Developments

RADMAP bits together into one platform, for real time radiation monitoring. It set up communication between the **Raspberry Pi-5**, the Gamma Spectrometer, the **GlobalSat BU-353N GPS Receiver**, several environmental sensors and the camera module. After that, the system synchronized radiation readings, *GPS coordinates, temperature, humidity, and a timestamp*, to make **2D heatmaps** and also allow **3D Digital Twin visualization**. Overall, this bundled architecture keeps the data handling running smoothly and it lays a scalable foundation for future improvements like AI based source localization and autonomous radiation monitoring, which is pretty handy.

Key Outcomes

- I sort of integrated all the hardware components into one unified backend, you know, it's like all the parts come together there.
- I enabled real time sensor data acquisition, and did the synchronization, so it all runs in step, basically as it happens.
- Support for 2D heatmap and also 3D Digital Twin visualization, kind a like a live model view in the background too.
- Developed a reliable data processing pipeline.

4.7 End-to-End Architecture of the GPS-Enabled Radiation Mapping System

A **3D prototype model** of the **RADMAP** system was put together to kind a mirror the final hardware setup, with the **Raspberry Pi-5**, **Gamma Spectrometer**, **GlobalSat BU-353N GPS Receiver**, a set of environmental sensors and also a camera module. Besides the

physical prototype, there was a **3D SLAM (Simultaneous Localization and Mapping)** model built, meant to form a digital version of the surrounding area, sort of like a living map. This **SLAM model** works as the base layer for the **3D Digital Twin**, allowing real time mapping and visualization, and in the future it supports radioactive source localization. Taken as a whole, the complete prototype gives a scalable ground platform for field use, autonomous navigation and more advanced radiation monitoring use cases, all connected in one approach.

So the proposed **RADMAP** architecture kind of sticks to a four-layer setup, made up of the IoT Sensor Layer, the Data Integration Layer, the Digital Twin Modelling Layer, and that Decision Intelligence Layer. In the IoT part it gathers real-time radiation readings, GPS coordinates, also temperature, humidity, and camera outputs using the Gamma Spectrometer, the **GlobalSat BU-353N GPS Receiver** , plus a **Raspberry Pi-5** and some environmental sensors. After that, the Data Integration Layer does the usual kind of work, it pre-processes, synchronizes, and stores what was collected, so later on it can be analysed properly. Then in the Digital Twin Modelling Layer it builds 2D radiation heatmaps along with a **3D Digital Twin** by pulling together GPS, **SLAM**, and live camera feedback, this should help with spatial visualization more and it also supports future radioactive source localization.

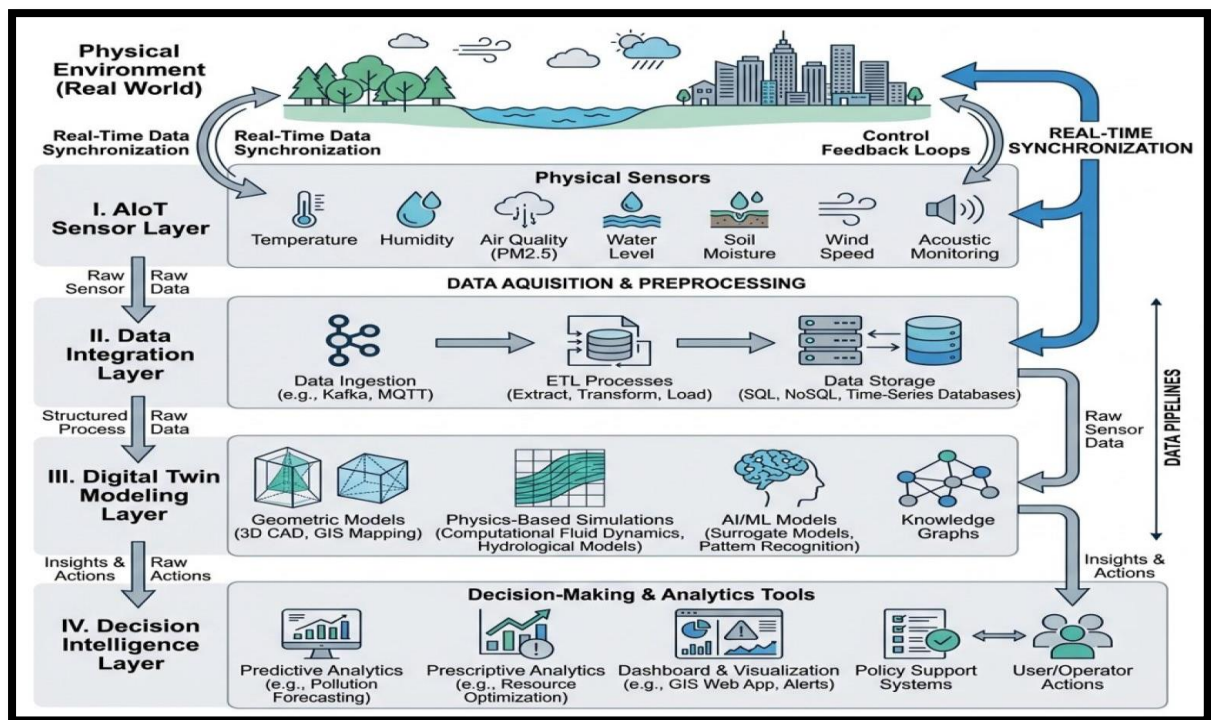


Figure 7: Overall Architecture of the RADMAP System layered-wise manner.

4.8 Prototype Design and SLAM Model

A 3D prototype model of the **RADMAP** system was made to kind of show the final hardware arrangement, it includes the **Raspberry Pi-5**, a Gamma Spectrometer, the **GlobalSat BU-353N GPS Receiver**, assorted environmental sensors and also the camera module. Beside the physical prototype, there was also a **3D SLAM model**, Simultaneous Localization and Mapping, worked on to build a digital version of the nearby area. This SLAM piece basically becomes the base for the **3D Digital Twin**, allowing real-time mapping, visualization, and later on radioactive source localization. Altogether the full prototype acts like a scalable setup for field deployment, autonomous navigation, and broader radiation monitoring use cases. And then at the end, the Decision Intelligence Layer presents the processed outputs through interactive dashboards, radiation analytics, and live visualization, which in practice supports radiation monitoring, decision making, and yeah, those future AI-based, autonomous mapping style scenarios too.



Figure 8: 3D Printed Prototype / CAD Model

RESULTS AND OUTCOME

5.1 Overview

The **RADMAP** project, kind of proved the feasibility of a GPS enabled real-time radiation mapping system using **Raspberry Pi-5**, plus a Gamma Spectrometer together with a GlobalSat **BU-353N GPS Receiver**, and then add environmental sensors and a live camera feed, kind of monitoring in parallel. During the internship we built a full backend pipeline to gather, handle, display, and study radiation data, basically all in real time.

One of the main achievements was really more about getting the 2D radiation heatmap to work properly, like mapping radiation intensity with GPS coordinates in a way that was accurate, even if it took some iterations. The system then showed radiation levels via moving, dynamic color gradients, so it was easier to spot the radiation hotspots fast. At the same time it still reported radiation counts, peak channels, and basic environmental parameters like temperature and humidity which mattered a lot.

And because conventional 2D mapping has that kind of limitation, a 3D Digital Twin was researched and then built by tying together GPS coordinates, SLAM based mapping, and live camera feedback. This ended up giving a more intuitive spatial representation of the radiation distribution, and in the long run it also set up the groundwork for future localization of radioactive sources, which is kind a the next step.

The **Gamma Spectrometer (CNSPEC MCA)** was successfully integrated with the Raspberry Pi-5 and experimentally validated using **Cobalt-60 (Co-60)** and **Cesium-137 (Cs-137)** radioactive sources. The detector accurately identified radiation counts, peak channels, and energy spectra, confirming the reliability of the developed system.

Real time backend architecture was put in place to sync radiation data, GPS information, environmental parameters and camera streams together into one processing pipeline. That backend keeps acquisition, storage and visualization efficient too, while also leaving the door open for future AI features, and even autonomous navigation.

On top of that, a *color grading algorithm* inspired by night vision imaging was developed. The goal was to overlay radiation intensity directly onto the live camera feed. Different

radiation levels were shown with a set of **color gradients**, it improved the real time visualization quite a lot, and honestly made situational awareness feel clearer in the moment.

Finally, a 3D prototype model of the **RADMAP** system was designed for future deployment, validating and testing the device. Based on the successful outcomes of the project, the work got funding approval from **Dr. Abhishek Yadav** and the Director, **Amity Institute of Nuclear Science and Technology (AINST)** for further enhancement, hardware integration , and next generation development.

5.2 Prototype Implementation

The final **RADMAP** prototype brings together the main hardware components into this kind of compact , portable setup. It does real time acquisition of radiation counts along with GPS coordinates, temperature, humidity and various environmental details. After that the gathered data go through the **Raspberry Pi-5** backend where they are processed to create synchronized radiation maps and to really help with Digital Twin visualization. Overall it kind of ties everything in one place, without being too bulky.



Figure 9: Final RADMAP Prototype Model

5.3 GPS Data Acquisition Results

The **GlobalSat ‘BU-353N’** GPS receiver managed to give continuous geo location and environmental telemetry while the system was in operation, more or less. The **GPS** data that was collected got synchronized with radiation readings so, in the end, accurate geospatial radiation maps could be generated. The **GlobalSat ‘BU-353N’** was then successfully put together with the **Raspberry Pi-5**, to capture real time geographical location info. Throughout operation the setup kept logging latitude, longitude, UTC time, satellite details, speed, and the positioning precision using HDOP and PDOP, via NMEA communication.

Parameter	Observed Value
GPS Receiver	GlobalSat BU-353N
Communication Port	COM8 (Prolific PL2303GC USB Serial)
Baud Rate	4800 bps
Date	18 th June, 2026
UTC Time	12:37:50
Latitude	28° 32.7622' N
Longitude	77° 19.9810' E
GPS Fix Status	3D Fix
Speed	1km/h
Direction (Course)	77.99°
Satellites in View	10
Satellites in Used	5
HDOP	1.57
PDOP	1.85
Data Protocol	NMEA 0183 (GPGGA, GPRMC, GPGSV, GPGSA)
Communication Interface	USB Serial (PL2303GC)

Table 3: Real-Time GPS Receiver Output (GlobalSat BU-353N)

6.4 Final Integrated System Metrics

The integrated system successfully synchronized radiation measurements with GPS coordinates and environmental parameters. These combined metrics were used to generate the 2D radiation heatmap and support the 3D Digital Twin visualization.

Parameter	Value
Source	Cobalt-60 (Co-60)
Device (Status)	LIVE
Elapsed Duration	38S
Total Counts	3,935
Radiation Intensity (Area)	3,935
Count Rate	103.6 CPS
Peak Channel	92
Max Counts (at Peak)	45

Table 4: Session Summary of Gamma Spectrometer

Channel Number	Count	Relative Prominence
92	45	Primary Photo peak
127	27	Secondary
148	24	Secondary
168	19	Minor
190	11	Minor
212	13	Minor
240	26	Secondary
274	13	Minor

Table 5: Detected Photo peaks (Channel-wise) of Gamma Spectrometer

CONCLUSION

The **RADMAP** internship project really, it turned out, showcased the creation of an integrated GPS enabled radiation mapping system. It can do real time radiation monitoring and also spatial visualization, sort of in one go. During the internship, i built the backend architecture end to end, by bringing together a **Raspberry Pi-5**, the **GlobalSat BU-353N GPS receiver**, a gamma spectrometer, plus environmental sensors and even a camera module into one unified setup. In practice the system could collect synchronized radiation counts, geographical coordinates, temperature, humidity, and a few other environmental inputs for on-going monitoring.

A, kind of notable achievement in this work was pushing forward a 2D radiation heatmap, kind a showing radiation intensity with color gradients and GPS tied measurements, it was more visual than before. To deal with the usual limitation you get from conventional 2D displays, research also went into **3D Digital Twin**, and it led to a real time, three dimensional radiation mapping framework. That setup brings together GPS data, live camera feedback, and radiation readings, in one system. Overall this method improves spatial awareness, and it lays groundwork—solid groundwork—for future work on radioactive source localization.

The Gamma Spectrometer was successfully integrated and, kind a experimentally validated, using **Cobalt-60 (Co-60)** and **Cesium-137 (Cs-137)** radioactive sources; this basically confirmed how accurate and reliable the radiation detection system is. Also, a night vision inspired color grading method was put in place so the radiation intensity could be shown in motion on the live camera view, which makes the radiation distribution easier to read during routine monitoring. Finally, a 3D prototype model was designed to reflect the final deployment setup, for the **RADMAP** system.

Overall, the internship gave some really valuable hands on practice in embedded systems, IoT, radiation detection, geospatial mapping, Digital Twin tech, backend system building, sensor integration, and, yeah also scientific research. The **RADMAP** prototype was implemented successfully, and the research team gave it a good evaluation, so basically the institution approved it and also offered funding support for more improvements. For what comes next, the plan is to plug in **SLAM** based navigation, AI assisted radioactive source localization, autonomous robot mobility, and more capable real time Digital Twin analytics,

so the **RADMAP** system can grow into a more complete intelligent radiation monitoring platform for research, industrial, and nuclear safety related applications.

6.1 Future Scope

The **RADMAP** project gives a solid groundwork for building an intelligent, autonomous radiation monitoring setup. Even if the current prototype is already doing real time radiation detection, GPS mapping, and Digital Twin visualization, in the next versions there is room to add a few more advanced things, just to boost overall capability and make real deployment easier.

Future improvements include:

- **AI driven radioactive source spotting:** build machine learning models that can figure out where a radioactive source is, and also estimate the position, based on the radiation intensity patterns that get detected, kind a like a fingerprint.
- **SLAM Integration for autonomous mapping:** we integrate Simultaneous Localization and Mapping (SLAM) approaches so the robot can figure out where it is while also building a map, specially in indoor places and in GPS-denied environments, with more accuracy.
- **Real-time 3D Digital Twin enhancement:** we boost the Digital Twin by bringing in LiDAR, depth cameras, and high resolution three dimensional reconstruction, so the spatial view looks clearer, more coherent, and easier to interpret.
- **Autonomous Robotic Navigation:** deploy the RADMAP system on ground robots that can drive on they own, or on unmanned vehicles, to do radiation monitoring in places that are hazardous, and really hard to reach.
- **Drone-Based Radiation Survey:** Extend the drone-based radiation survey platform to UAVs, you know, basically drones, so they can do rapid aerial radiation checking during nuclear emergencies and also for environmental monitoring, rather than waiting on ground teams.
- **Cloud based monitoring dashboard:** you build a cloud platform that's used for remote oversight, data storage and real time visualization too, plus multi user access that works from anywhere.

- **Predictive Radiation Analytics:** bring in Artificial Intelligence to estimate how radiation spread will, identify odd patterns in advance, and also create early warning alerts, sooner than later.
- **Multi-Sensor Fusion:** mix radiation readings with LiDAR, thermal imagery, weather details and environmental sensors so detection stays steadier, and the situational awareness feels more clear.
- **Real-Time Emergency Response System:** craft automated alarm mechanisms that notify the operators whenever radiation levels go above the already set safety thresholds, without delay.
- **Field Deployment and Large-Scale Validation:** run prolonged outdoor trials in industrial, research and nuclear settings to judge system reliability, its resilience and long term operational performance properly.